

HIDDEN RADIOLOGY

ANA FAHIM

CHEST X-RAY MASTERY

Fundamental Radiographic Principles,
Systematic Interpretation & Clinical Pathology

See Beyond The Image

Volume 1 • English Edition

About This Book

This handbook is an educational reference designed to make chest radiography more systematic, practical, and easier to review. It combines technical image-quality assessment, a structured reading sequence, common radiographic signs, technical pitfalls, and rapid-reference checklists.

How to Use This Book

- Start with technical quality before interpreting pathology.
- Use the ABCDEFGHI sequence consistently to reduce search errors.
- Compare both sides whenever possible.
- Use checklists as a final safety net.

Contents

1. Chapter 1 — Technical Standards & Image Quality Assessment
2. Chapter 2 — The ABCDEFGHI Systematic Interpretation Method
3. Chapter 3 — Core Chest Pathologies & Radiographic Signs
4. Chapter 4 — Artifacts, Technical Pitfalls & Positioning Errors
5. Chapter 5 — Lines, Tubes, Devices & Rapid Reference Checklists

Chapter 1

Technical Standards & Image Quality Assessment

Before You Interpret

Before interpreting a chest radiograph, first decide whether the image is technically adequate for the clinical question. The R.I.P. framework—Rotation, Inspiration, and Penetration—provides a rapid starting point. Also assess coverage, laterality, and collimation.

PA vs. AP

A standard routine chest examination commonly uses an erect PA and left lateral projection when clinically appropriate. AP imaging may be substituted when the patient's condition prevents standard positioning. AP projection can increase apparent cardiac magnification. ■cite■turn0search11■turn0search3■

The R.I.P. Check

Rotation: assess clavicular symmetry. Inspiration: judge lung expansion in context; about 9–10 posterior ribs is a common teaching criterion. Penetration: ensure the retrocardiac lung and lower thoracic spine are sufficiently visible. Coverage and collimation should include the apices and costophrenic sulci when possible. ■cite■turn0search11■turn0search6■

Chapter 2

The ABCDEFGHI Systematic Interpretation Method

A — Airway

Assess tracheal position, airway caliber, and the main bronchi. Consider whether deviation is pushed away or pulled toward a process.

B — Bones & Soft Tissues

Review ribs, clavicles, shoulders, visible spine, and soft tissues for fractures, destructive lesions, subcutaneous emphysema, or other abnormalities.

C — Cardiac Silhouette & Great Vessels

Assess heart size and contour with projection in mind. A cardiothoracic ratio is most meaningful on an appropriate PA projection and should not be overinterpreted on AP portable images.

D — Diaphragm

Compare hemidiaphragm position, assess the costophrenic and cardiophrenic angles, and look beneath the diaphragm for unexpected free air when visible.

E — Effusion & Pleura

Inspect pleural margins and costophrenic angles. Look for pleural fluid, pneumothorax, pleural thickening, or other pleural abnormalities.

F — Fields / Lungs

Compare upper, middle, and lower zones bilaterally. Search for focal air-space opacity, interstitial pattern, nodules, volume loss, or abnormal lucency.

G — Great Vessels & Hila

Assess hilar size, symmetry, vascularity, and the visible aortic contour. Remember that normal hilar position and prominence vary.

H — Hila & Mediastinum

Review mediastinal contours and widths while accounting for projection and rotation. Look for abnormal widening, displacement, or mass effect.

I — Impression, Lines & Devices

Confirm lines and tubes when present, synthesize the major findings, and formulate a concise impression. The final clinical interpretation should integrate history and comparison studies when available.

Chapter 3

Core Chest Pathologies & Radiographic Signs

3.1 Pneumonia & Alveolar Consolidation

Typical air-space consolidation may appear as increased opacity with air bronchograms. The silhouette sign occurs when an interface between structures of similar radiographic density is lost. Loss of the right heart border → consider right middle lobe involvement. Loss of the right hemidiaphragm → consider right lower lobe involvement. Loss of the left heart border → consider lingular involvement.

3.2 Pneumothorax

Look for a visible visceral pleural line with absent peripheral lung markings beyond it. In a supine patient, the appearance can differ from the classic upright pattern. Always assess for signs of tension physiology in the appropriate clinical context. Tension pneumothorax is a medical emergency and should not be delayed for textbook radiographic criteria when clinical instability is present.

3.3 Pleural Effusion

On an erect PA radiograph, pleural fluid may blunt the lateral costophrenic angle and produce a meniscus. Small effusions can be subtle, and positioning affects sensitivity. Large effusions may opacify much of a hemithorax and can shift the mediastinum away when mass effect is present. Do not rely on a single fixed milliliter threshold; radiographic visibility varies with projection and patient position.

3.4 Congestive Heart Failure & Pulmonary Edema

Radiographic findings may include vascular redistribution, interstitial edema, peribronchial cuffing, septal lines, pleural effusions, and bilateral air-space opacities. Findings should be interpreted with the clinical picture. Stage-based teaching can be useful, but radiographic signs do not map perfectly to a single pressure threshold. Cardiomegaly and pulmonary edema may coexist, but one does not automatically prove the other.

Chapter 4

Artifacts, Technical Pitfalls & Positioning Errors

Motion & Breathing Artifacts

Motion can blur vascular and diaphragmatic margins and reduce diagnostic confidence.

Rotation Artifacts

Rotation can create apparent differences in lung density, hilar position, and mediastinal width.

External / Skin Artifacts

Clothing, hair, skin folds, monitoring equipment, and other external objects can project over the lungs.

Portable Chest Pitfalls

Portable AP images require particular caution with cardiac size, mediastinal contours, rotation, incomplete inspiration, and supine pleural findings. ■cite■turn0search12■

Chapter 5

Lines, Tubes, Devices & Rapid Reference Checklists

Device Positioning — Teaching Reference

ETT: assess tip relative to the carina and account for head position. CVC: trace the catheter and assess the tip according to route and institutional standard. Enteric tube: trace continuously and confirm the distal course is appropriate.

Rapid Clinical Pre-Flight Checklist

Patient identity and laterality • Projection • Coverage • Rotation • Inspiration • Exposure • Airway • Bones/soft tissues • Heart/mediastinum • Lungs/hila/pleura/diaphragms • Lines/tubes • Prior imaging • Final impression.

Production Notes & Medical Review Status

The core teaching structure of the supplied manuscript has been retained, while rigid statements were reframed where projection, positioning, or clinical context can change interpretation. External verification for this production pass used ACR chest-radiography practice-parameter material and Royal College of Radiologists image-quality criteria. ACR describes its parameters as educational tools rather than inflexible rules. ■cite■turn0search0■turn0search3■

Clinical Disclaimer

This book is for educational purposes only. It does not replace formal clinical guidelines, institutional protocols, supervision, or professional medical judgment.